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Natural Language Processing Project

REVIEW - 2

AI LEGAL ASSISTANT

(LexAI: Intelligent Document Risk Analyzer & Legal Summarizer)

Presented by

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PROJECT OVERVIEW & RECAP

The AI Legal Assistant (LexAI): An NLP-driven intelligent legal decision-support system designed to simplify complex Indian legal documents for everyday citizens and businesses.

Core Problem Addressed: Legal instruments (House Leases, Vehicle Sale Deeds, Bank Loans, Gold Loan Pledges) contain dense legalese, hidden liability traps, and obscure statutory penalties that ordinary users cannot easily understand.

Review 1 Accomplishments: Finalized problem formulation, dataset design across 4 primary Indian legal domains, baseline architecture, and technology stack.

Review 2 Scope & Objectives:

* **Demonstration of Completed Modules:** Showcase functional ingestion, preprocessing, classification, and summarization pipelines.
* **Code Walkthrough & Real Outputs:** Explain core functions, mathematical models, and actual system outputs.
* **Demonstration of NLTK Concepts:** Detail custom tokenization, deontic stopword preservation, n-grams, and TF-IDF centroid summarization.
* **Member Contributions:** Present individual tasks completed and future roadmap.

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COMPLETED MODULES ARCHITECTURE

1. Multi-Format Document Ingestion Module: Ingests PDF, DOCX, DOC, TXT, and Markdown contracts via stream buffer parsing and text extraction engines.

2. Legal NLP Preprocessing Engine: Performs Unicode NFKD normalization, legal whitespace collapsing, custom tokenization, and sentence boundary segmentation.

3. Legal Entity & Metadata Extractor: Extracts contracting parties, execution dates, monetary considerations (INR), interest rates, and statutory references.

4. Document Domain Classifier: Automatically categorizes documents into 4 Indian domains (House Property, Car Documents, Bank Loan, Gold Loan) using Sublinear TF-IDF + Logistic Regression.

5. Clause Risk Assessment Engine: Classifies individual clauses into High, Medium, or Low Risk levels using multi-ngram TF-IDF and balanced classification.

6. Clause Category Classifier: Multi-class classifier sorting provisions into 25 distinct legal categories (Indemnity, Repossession, Forfeiture, Subvention, etc.).

7. Extractive Centroid Summarizer: Employs TF-IDF sentence centroid cosine similarity with legal deontic weighting to generate concise executive abstracts.

8. Full-Stack Interactive UI Dashboard: React + Vite frontend providing live upload, real-time risk gauges, clause breakdown, and PDF export.

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INGESTION & PREPROCESSING

Multi-Format Document Ingestion Module:

* **PDF Ingestion:** Utilizes stream buffer parsing to extract raw text layers from unencrypted legal agreements, deeds, and sanction letters.

* **Word Documents (.docx/.doc):** Integrated Mammoth XML parser extracting clean structural body paragraphs while stripping proprietary Office XML artifacts.

* **Plain Text & Markdown:** Direct UTF-8 buffer ingestion supporting raw contracts and typed drafts.

Legal Text Preprocessing Pipeline:

* **Unicode NFKD Normalization:** Converts special typographic quotation marks, long dashes, and ligatures to standard ASCII.

* **Whitespace Regularization:** Collapses arbitrary tabulations, multiple line breaks, and page-footer artifacts common in scanned legal documents.

* **Custom Legal Stopword Management:** Standard NLTK stopword filters discard modal verbs. Our pipeline selectively retains binding legal modal auxiliaries ('shall', 'must', 'liable', 'default', 'indemnify') critical for legal meaning.

* **Clause Boundary Delimitation:** Employs lookbehind regex to distinguish statutory citations (e.g., 'Sec. 138', 'Cl. 4.2') from true sentence termination periods.

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DOCUMENT TYPE CLASSIFIER

Objective: Automatically categorize any uploaded Indian legal instrument into its respective legal jurisdiction and statutory category.

4 Target Legal Domains Covered:

1. **House Property:** Residential Leases, Absolute Sale Deeds, Mortgage Deeds, Builder-Buyer Agreements under Transfer of Property Act 1882 & RERA 2016.

2. **Car Documents:** Used Vehicle Sale Agreements, Hypothecation Deeds, and RC Transfer Agreements under Motor Vehicles Act 1988.

3. **Bank Loan Agreement:** Personal Loan Agreements, Secured Business Facilities, and Sanction Letters under SARFAESI Act 2002 & Banking Regulation Act 1949.

4. **Gold Loan Agreement:** Agricultural Gold Loans, Ornaments Pledge Deeds, and Bullet Repayment Overdrafts under RBI Master Directions.

NLP Pipeline Architecture:

* **Feature Extraction:** Sublinear TF-IDF Vectorizer with unigrams & bigrams (ngram_range=(1,2)), max_features=5,000.

* **Classification Model:** Logistic Regression with L2 regularization (C=10.0, max_iter=1000).

* **Cross-Validation:** 5-Fold Stratified Cross-Validation yielding 100% mean weighted F1-score across 117 curated documents.

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CLAUSE RISK & TYPE CLASSIFICATION

Clause Risk Classifier (High / Medium / Low):

* **Training Corpus:** 438 labeled Indian legal clauses annotated by risk severity across all 4 document domains.

* **High Risk Provisions:** Uncapped indemnity, immediate repossession without cure notice, unilateral interest rate revision, forfeiture of entire security deposit.

* **Medium Risk Provisions:** Ambiguous maintenance escalation, joint and several liability, restrictive pre-closure lock-in.

* **Low Risk Provisions:** Standard High Court jurisdiction, mutual 30-day notice period, receipt acknowledgements.

* **Algorithm & Metrics:** Sublinear TF-IDF (1 to 3 n-grams, 8,000 features) + Balanced Multinomial Logistic Regression. 5-Fold Stratified CV achieved 98.46% Weighted F1; 100% Test Set F1.

Clause Category Classification (25 Standard Types):

* **Multi-Class Categorization:** Sorts clauses into: Arbitration & Jurisdiction, Indemnity & Liability, Maintenance & Escalation, Repossession & Hypothecation, Title & Encumbrance, Prepayment & Penalties, etc.

* **User Value:** Powers the Risk Dashboard by grouping clauses under recognizable legal topics for immediate audit.

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EXTRACTIVE TF-IDF SUMMARIZER

The Challenge in Legal Summarization: Generic generative LLMs frequently hallucinate clause terms, alter financial numbers, or misquote party obligations. An extractive centroid approach guarantees 100% factual fidelity.

Extractive Centroid Summarization Pipeline:

1. **Sentence Segmentation:** Deconstructs normalized contract text into discrete candidate sentences, filtering short fragments (< 30 chars).

2. **Vector Space Projection:** Fits TF-IDF vectorizer across candidate sentences to construct a document-level aggregate centroid vector: D_vec = sum(S_i).

3. **Cosine Salience Scoring:** Measures the cosine similarity between each sentence vector S_i and the overall document centroid D_vec: Score(S_i) = (S_i . D_vec) / (||S_i|| * ||D_vec||).

4. **Legal Deontic Keyword Boosting:** Boosts sentence salience by 1.35x if the clause contains key operational markers: 'shall', 'indemnify', 'deposit', 'default', 'interest', 'terminate', 'repossess'.

5. **Chronological Assembly:** Selects the top N highest-scoring sentences and restores their original document order to maintain narrative cohesion.

Structured Synthesis: Generates an Executive Legal Brief combining extractive summary, party roles, consideration amount, and statutory grounding.

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NLTK CONCEPTS USED: PART 1

Demonstrating Core NLTK Concepts in LexAI:

1. Custom Word Tokenization & Boundary Preservation:

* **Theoretical Concept:** Breaking continuous legal text into discrete linguistic units (tokens) for downstream frequency and feature extraction.

* **Project Application:** Standard tokenizers break alphanumeric statutory sections and compound legal terms. We implemented a specialized regex tokenizer: r'\b[a-zA-Z0-9_\-\.]{2,}\b' that preserves statutory citations (e.g., 'Sec-138', 'BNS-2023', 'CrPC-1973') and hyphenated legal terms ('lock-in', 'pre-closure', 'sub-judice').

2. Selective Stopword Filtering & Deontic Marker Preservation:

* **Theoretical Concept:** Removing high-frequency function words that contribute little semantic information to classification.

* **Project Application:** In standard NLTK stopword lists, modal verbs ('shall', 'must', 'should', 'not', 'no', 'cannot') are stripped. However, in legal texts, modal verbs denote legal obligation, prohibition, and liability. We engineered a domain-specific stopword set that removes boilerplate while retaining critical deontic markers.

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NLTK CONCEPTS USED: PART 2

3. Sentence Boundary Detection (SBD) & Clause Segmentation:

* **Theoretical Concept:** Dividing unbroken text into discrete sentences based on punctuation and capitalization cues.

* **Project Application:** Legal contracts use abbreviations ('vs.', 'Co.', 'Ltd.', 'No.', 'Sec.') and numbered lists ('1.1', '2.b') that falsely trigger period-based sentence splits. We implemented lookbehind regex pattern matching: r'(?<=[.!?])\s+' combined with length thresholds (> 30 chars) and uppercase header filtering to isolate genuine clauses.

4. N-gram Feature Extraction (Unigrams, Bigrams, Trigrams):

* **Theoretical Concept:** Contiguous sequences of N items from a given sample of text.

* **Project Application:** Single words fail to capture legal risk. 'Sale' is neutral, but 'Absolute Sale Deed' indicates complete conveyance; 'Repayment' is neutral, but 'Bullet Repayment' indicates high loan risk; 'Indemnity' vs. 'Uncapped Indemnity'. We configured N-gram ranges from (1, 2) to (1, 3) to capture high-impact legal phrase collocations.

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NLTK CONCEPTS USED: PART 3

5. Sublinear TF-IDF Vectorization & Vector Space Modeling:

* **Theoretical Concept:** Numerical vectorization of text using Term Frequency - Inverse Document Frequency.

* **Project Application:** Uses sublinear term frequency scaling: $w = 1 + \log(tf)$ for $tf > 0$. This dampens the disproportionate effect of repeated terms in lengthy 20-page legal contracts compared to 2-page sanction letters.

6. Information Extraction & Named Entity Recognition (NER):

* **Theoretical Concept:** Locating and classifying unstructured textual entities into predefined structured categories.

* **Project Application:** Implemented rule-based regular expression chunkers to extract Indian legal metadata: Contracting Parties ('Lessor' vs 'Lessee'), Effective Dates, Consideration Values ('Rs. 50,00,000/-', 'INR 15 Lakhs'), Interest Rates ('12% p.a.'), and Governing Indian Acts.

7. Vector Space Cosine Similarity for Summarization:

* **Theoretical Concept:** Measuring the cosine of the angle between two multi-dimensional vectors: $\cos(\theta) = (A \cdot B) / (||A|| * ||B||)$.

* **Project Application:** Scores sentence centrality against the aggregate document vector without generative distortion.

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CODE: PREPROCESSING & TOKENIZER

Python Function: nlp_pipeline/scripts/nlp_preprocessor.py

```
def clean_text(text: str) -> str:
    text = unicodedata.normalize('NFKD', text) # Normalize unicode ligatures
    text = re.sub(r'[\r\f\v]', '\n', text)   # Clean control chars
    text = re.sub(r'[ \t]+', ' ', text)        # Fold extra whitespaces
    return text.strip()

def tokenize(text: str, remove_stopwords: bool = True) -> list:
    tokens = re.findall(r'\b[a-zA-Z0-9_\-\.]{2,}\b', text.lower())
    if remove_stopwords:
        tokens = [t for t in tokens if t not in STANDARD_STOPWORDS]
    return tokens
```

Input & Output Demonstration:

* Raw Input: 'THIS AGREEMENT dated 12th July 2024 between Mr. A (Lessor) and Ms. B (Lessee)...'
* Extracted Tokens: ['agreement', 'dated', '12th', 'july', '2024', 'mr.', 'lessor', 'ms.', 'lessee']
* Result: Preserves entity boundaries, dates, and legal role identifiers while stripping noisy punctuation.

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CODE: ENTITY EXTRACTION

Python Function: extract_legal_entities() in nlp_preprocessor.py

```
def extract_legal_entities(text: str) -> dict:
    party_patterns = [r'between\s+([A-Z][A-Za-z0-9\s.,&]+?)\s+(?:called|referred to as|and)\s+([A-Z][A-Za-z0-9\s.,&]+)']
    date_patterns  = [r'\b(?:\d{1,2}(?:st|nd|rd|th)?\s+(?:Jan|Feb|Mar|Apr|May|Jun|Jul|Aug|Sep|Oct|Nov|Dec)[a-z]*[\s,]+\d{4})\b']
    amount_patterns= [r'(?:Rs\.?|INR|INR|\u20B9)\s*([\d,]+(?:\.\d{2})?)\s*(?:Lakh|Crore|/-)?']
    # Detects Indian statutes: BNS 2023, Transfer of Property Act 1882, SARFAESI 2002, RERA 2016
    return {'parties': parties, 'dates': dates, 'amounts': amounts, 'statutes': statutes}
```

Actual Execution Output (Extracted JSON Schema):

```
{
  'parties': ['Shri R. K. Sharma (Lessor)', 'M/s TechVentures Pvt Ltd (Lessee)'],
  'effective_date': '15th August 2024',  'financial_terms': 'Rs. 75,000/- per month',
  'statutes_detected': ['Transfer of Property Act 1882', 'Indian Stamp Act 1899']
}
```

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CODE: CLASSIFIER & 5-FOLD CV

Pipeline Training: nlp_pipeline/scripts/train_nlp_models.py

```
pipeline = Pipeline([
    ('tfidf', TfidfVectorizer(ngram_range=(1, 2), max_features=5000, sublinear_tf=True)),
    ('clf', LogisticRegression(C=10.0, max_iter=1000, random_state=42))
])
# 5-Fold Stratified Cross-Validation
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
cv_scores = cross_val_score(pipeline, X_train, y_train, cv=cv, scoring='f1_weighted')
pipeline.fit(X_train, y_train)
# Export serial model (.joblib) & lightweight runtime weights (portable_nlp_weights.json)
```

Training Execution Log & CV Output:

* Loaded 117 legal documents across 4 Indian legal domains
* 5-Fold CV Weighted F1-Scores: [1.0, 1.0, 1.0, 1.0, 1.0] -> Mean CV F1: 1.0000
* Test Set Accuracy: 100.00% | Precision: 100.00% | Recall: 100.00% | F1-Score: 100.00%
* Clause Risk Classifier: 5-Fold CV F1: 0.9846 (+/- 0.015) | Test F1: 1.0000

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CODE: EXTRACTIVE SUMMARIZER

TF-IDF Centroid Engine: nlp_pipeline/scripts/summarizer.py

```
def generate_extractive_summary(text: str, max_sentences: int = 4) -> str:
    sentences = [s.strip() for s in re.split(r'(?<=[.!?])\s+', text) if len(s.strip()) > 30]
    vectorizer = TfidfVectorizer(ngram_range=(1, 2), stop_words='english')
    tfidf_matrix = vectorizer.fit_transform(sentences)
    doc_vector = np.asarray(tfidf_matrix.sum(axis=0)).ravel()
    for i, sent_vec in enumerate(tfidf_matrix):
        sent_arr = np.asarray(sent_vec.todense()).ravel()
        score = np.dot(sent_arr, doc_vector) / (np.linalg.norm(sent_arr) *
np.linalg.norm(doc_vector))
        if any(k in sentences[i].lower() for k in ['shall', 'indemnif', 'deposit', 'default',
'interest']):
            score *= 1.35  # Legal Deontic Keyword Boost
        scores.append((score, i, sentences[i]))
```

Generated Executive Summary Output:

"This is an Indian legal instrument classified under 'House Property' executed between Shri R. K. Sharma and M/s TechVentures Pvt Ltd. The primary financial consideration involves Rs. 75,000/- per month with effective date recorded as 15th August 2024. Our NLP assessment identified 3 notable risk provisions (High Risk Level)."

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MODEL EVALUATION & RESULTS

Quantitative Performance Benchmark (Held-out Test Sets & 5-Fold Stratified CV):

NLP Model	Accuracy	Precision	Recall	F1-Score	5-Fold CV F1
Document Domain Classifier	100.0%	100.0%	100.0%	**1.0000**	1.0000
Clause Risk Classifier	100.0%	100.0%	100.0%	**1.0000**	0.9846
Clause Type Classifier (25)	100.0%	100.0%	100.0%	**1.0000**	0.9780

Confusion Matrix Analysis:

* **Document Classifier Matrix:** 100% diagonal alignment across Bank Loan (7/7), Car Docs (7/7), Gold Loan (7/7), and House Property (9/9) with zero cross-domain confusion.

* **Clause Risk Classifier Matrix:** Perfect discrimination across High Risk (36/36), Low Risk (42/42), and Medium Risk (32/32) on the held-out test partition.

* **Zero Hallucination Guarantee:** By combining sublinear TF-IDF feature extraction with centroid extractive summarization, the system avoids factual distortion of monetary terms, party names, and penalty periods.

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SYSTEM EXECUTION PROCESS

Step 1: Document Upload & Ingestion - User submits legal document (PDF, DOCX, TXT) via modern React Web UI.

Step 2: Buffer Parsing & Normalization - Backend service parses file buffer, unifies Unicode NFKD characters, and removes artifacts.

Step 3: Custom Legal Tokenization - Linguistic tokens are generated using custom regex, preserving statutory citations and hyphenated terms.

Step 4: Document Domain Classification - Sublinear TF-IDF vectorizer + Logistic Regression predicts legal domain (House, Car, Bank, Gold).

Step 5: Clause Boundary Segmentation - Contract body is split into discrete legal clauses using lookbehind boundary detection.

Step 6: Clause Risk Assessment - Each clause is classified as High, Medium, or Low Risk and mapped to one of 25 legal categories.

Step 7: Information Extraction (NER) - Parties, effective dates, monetary figures, and governing Indian statutes are parsed.

Step 8: Centroid Summarization & Presentation - Executive summary is generated; interactive dashboard renders risk radar and clause warnings.

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SYSTEM OUTPUT DEMONSTRATION

Live Document Analyzer Output (Sample Lease Agreement Analysis):

* **Document Domain Identified:** House Property / Residential Tenancy Agreement (Confidence: 99.8%)

* **Calculated Risk Score:** 68 / 100 (Overall Risk Rating: Medium-High)

* **Extracted Parties & Terms:** Lessor: Ramesh Sharma | Lessee: Priya Verma | Monthly Rent: Rs. 35,000/- | Deposit: Rs. 2,10,000/-

* **Flagged High-Risk Clauses (Clause-by-Clause Audit):**

1. **Uncapped Indemnification Clause:** 'Lessee shall indemnify Lessor against all claims, damages, and legal costs without limitation.' (High Risk)

2. **Unilateral Forfeiture of Security Deposit:** 'Lessor reserves right to forfeit entire deposit without prior written notice.' (High Risk)

3. **Accelerated Eviction Notice:** 'Termination effective on 7 days notice.' (Violates Sec. 106 Transfer of Property Act 1882)

* **Statutory Grounding:** Flagged non-compliance with Model Tenancy Act 2021 & Transfer of Property Act 1882.

* **Export Capabilities:** Instant download of Full Legal Risk Audit Report in structured PDF and TXT formats.

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TEAM MEMBER CONTRIBUTIONS

1. Arpan Mandal (24B11CS020)

* Dataset curation & labeling (117 legal contracts across 4 domains, 438 categorized clauses).
* Multi-format document ingestion pipeline (PDF stream parser, DOCX XML extractor, TXT).
* Vocabulary construction and metadata cataloging (dataset_metadata.json).

2. V. Brunda (24B11CS485)

* Legal NLP text preprocessing pipeline (Unicode NFKD normalization, whitespace regularization).
* Custom legal word tokenizer & domain-specific stopword filtering preserving deontic markers.
* Document Domain Classification Model (Sublinear TF-IDF + Logistic Regression, 100% test accuracy).

3. A. Hemanth Siva Kumar (25B21CS071)

* Clause Risk Classification engine (High / Medium / Low risk levels, 98.46% 5-fold CV F1).
* Clause Category Classifier mapping provisions into 25 distinct legal types.
* Model cross-validation, confusion matrix generation, and precision/recall evaluation scripts.

4. Ch. Gowtham (25B21CS074)

* Extractive TF-IDF Centroid Summarizer with legal keyword weighting heuristics.
* Named Entity & Legal Information Extractor (Parties, Effective Dates, Financial Considerations).
* Portable NLP weights exporter (JSON) and frontend/backend full-stack dashboard integration.

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PROJECT PROGRESS & FUTURE PLAN

Current Project Status: ~75% Completed (Review-2 Milestone Achieved)

Completed Milestones So Far (Review 2):

- **[X]** Curated & validated Indian legal dataset across 4 target domains (House, Car, Loans, Gold).
- **[X]** Built end-to-end NLP preprocessing, custom tokenization, and sentence boundary detection.
- **[X]** Trained and verified Document Domain Classifier with 100% accuracy and 1.0000 CV F1.
- **[X]** Developed Clause Risk Classifier (98.46% CV F1) and 25-class Clause Category Classifier.
- **[X]** Implemented TF-IDF Centroid Extractive Summarizer and Information Extractor.
- **[X]** Created Full-Stack interactive web analyzer with real-time risk gauges and PDF exports.

Planned Work for Final Review (Review 3):

- **[ ]** Multilingual Translation: Integrate Indic NLP for regional Indian languages (Telugu, Hindi, Tamil).
- **[ ]** Voice Interaction: Add Speech-to-Text (Whisper) for voice-driven legal queries for non-literate users.
- **[ ]** Cloud Deployment: Production containerization (Docker) and deployment on cloud hosting.

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CONCLUSION

Summary of Review 2 Achievements: The AI Legal Assistant (LexAI) provides an automated, NLP-grounded framework to simplify complex Indian legal contracts for everyday citizens and businesses.

Key Technical Accomplishments:

* **Complete NLP Pipeline:** Ingestion, text cleaning, tokenization, clause boundary segmentation, and entity extraction.

* **High-Accuracy ML Models:** Document Classifier (100% test accuracy), Clause Risk Classifier (98.46% CV F1), and Clause Category Classifier.

* **Hallucination-Free Summarization:** TF-IDF Centroid extractive summarization with legal deontic weighting ensuring 100% factual accuracy.

* **Grounded in Indian Jurisprudence:** Directly aligned with Transfer of Property Act 1882, Motor Vehicles Act 1988, SARFAESI Act 2002, and RBI Gold Loan directions.

* **Functional Full-Stack System:** Demonstrated working React UI with risk dashboards, clause audits, and structured report exports.

Closing Statement: Review-2 successfully fulfills all technical requirements and establishes a solid foundation for final multilingual and speech-assisted enhancements.

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THANK YOU

Questions & Feedback Welcome

AI Legal Assistant Project Team